

Self-Emergent Fourier Cymatics: Entropic Eigenmodes out of Chaos

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Abstract

From a curvature-regularized wave-confinement Lagrangian $\mathcal{L}_{\text{WCT}}$, we derive (i) the fourth-order Euler–Lagrange dynamics and (ii) a parabolic surrogate with strict Lyapunov descent $E_0[\psi] \downarrow$. Starting from random 0D seeds, a semi-implicit IMEX scheme (with Armijo guard) yields monotone E_0 and spectral-entropy $H_k \downarrow$, contracting power to a finite annulus and then to a dynamically selected finite support K^* of stationary peaks that coincide with Laplacian eigenmodes in bounded domains. A center-manifold reduction closes to a Swift–Hohenberg selector explaining finite-band locking. Linearization of the full Euler–Lagrange system about the locked state produces a positive curvature-induced gap Δ_* with dispersion $\omega_j^2 = c^2 \lambda_j + \Delta_*$, setting $m_{\text{eff}} = \hbar \sqrt{\Delta_*}/c^2$. We provide proofs of well-posedness/coercivity and a complete reproducibility package (parameters, seeds, CLI, code+data), establishing the tested chain $0\text{D} \Rightarrow E_0 \downarrow, H_k \downarrow \Rightarrow K^* \Rightarrow \{\phi_j\} \Rightarrow \Delta_* \Rightarrow m_{\text{eff}}$.

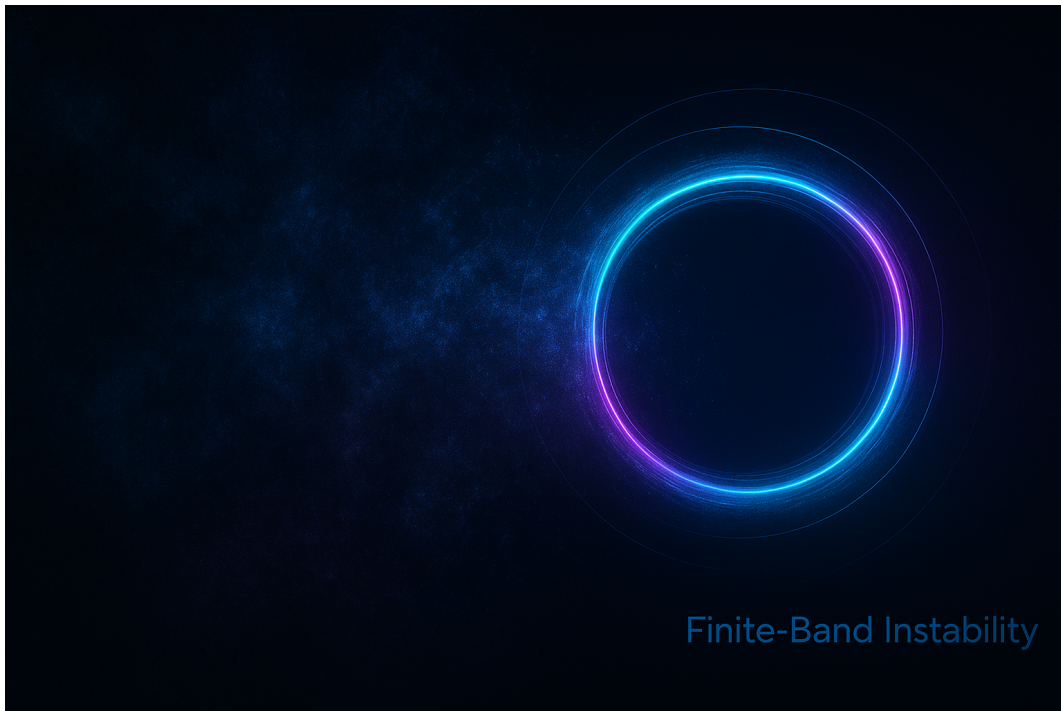


Figure 1: Cover illustration: Self-Emergent Fourier Cymatics — entropic eigenmodes out of chaos.

Canonical Wave Confinement Lagrangian

Definitions (fixed first). Let $\psi : \Omega \times \mathbb{R} \rightarrow \mathbb{C}$, $\square := \partial_t^2 - \Delta$, and

$$D(\psi, \bar{\psi}) := \psi + \epsilon e^{-\alpha|\psi|^2}, \quad F := D^{-1}.$$

Parameters: $\epsilon > 0$ (regularizer), $\alpha > 0$ (feedback strength), $\kappa, \theta \geq 0$ (curvature weights), and $\gamma \in \mathbb{R}$ (mixing). Potential $V : [0, \infty) \rightarrow \mathbb{R}$ is C^2 with $V'(s)$ denoting $\frac{dV}{ds}$.

Single canonical form (one label).

$$\boxed{\mathcal{L}_{\text{WCT}} = |\partial_\mu \psi|^2 - V(|\psi|^2) + \kappa \left(\frac{\square \psi}{D} \right)^2 + \theta \left(\frac{\Delta \psi}{D} \right)^2 + \gamma \left(\frac{\square \psi}{D} \right) \left(\frac{\Delta \psi}{D} \right)} \quad (1)$$

Coercivity near $\psi = 0$ is ensured by $\kappa, \theta \geq 0$ and the mixed-term bound $|\gamma| \leq 2\sqrt{\kappa\theta}$ (completing the square in the curvature sector). The $\epsilon > 0$ limit is taken *after* variation to avoid singularities at $\psi = 0$.

Euler–Lagrange (symbolic, with operators defined). Define the normalized curvatures

$$Q := \frac{\square \psi}{D}, \quad R := \frac{\Delta \psi}{D}.$$

Then the stationary action $\delta S = 0$ gives the fourth-order EL equation

$$-\square \psi - V'(|\psi|^2) \psi + 2\kappa \mathcal{D}_\square[\psi] + 2\theta \mathcal{D}_\Delta[\psi] + \gamma \mathcal{D}_{\square\Delta}[\psi] = 0, \quad (2)$$

where the curvature-feedback operators collect the exact variational terms

$$\mathcal{D}_\square[\psi] := \square(F^2 \square \psi) + \left(\partial_{\bar{\psi}} F^2 \right) (\square \psi)^2, \quad \mathcal{D}_\Delta[\psi] := \Delta(F^2 \Delta \psi) + \left(\partial_{\bar{\psi}} F^2 \right) (\Delta \psi)^2,$$

$$\mathcal{D}_{\square\Delta}[\psi] := \square(F^2 \Delta \psi) + \Delta(F^2 \square \psi) + 2 \left(\partial_{\bar{\psi}} F^2 \right) (\square \psi) (\Delta \psi),$$

with $\partial_{\bar{\psi}} F^2 = 2F \partial_{\bar{\psi}} F = 2F^2 \frac{\alpha \epsilon e^{-\alpha|\psi|^2} \psi}{D}$. (Full Gateaux-derivative steps are provided in App. A.)

Parabolic surrogate (pointer). Near finite- k onset, a gradient-flow surrogate (e.g. Swift–Hohenberg/CGL form) is introduced in Sec. ?? for numerics, chosen so that the discrete scheme obeys a strict Lyapunov descent $E \downarrow$ while reproducing the measured dispersion band.

Euler-Lagrange Field Equations

Variation of (1) yields the Euler–Lagrange PDE schematically as:

$$\partial_t^2 \psi - \nabla^2 \psi + V'(|\psi|^2) \psi + \kappa \delta_\psi \left[\frac{\square \psi}{\psi + \epsilon e^{-\alpha |\psi|^2}} \right] + \dots = 0 \quad (3)$$

For numerical stability in the simulation, we employ a parabolic evolution form equivalent to gradient descent in the associated energy functional:

$$\partial_t \psi = D \nabla^2 \psi - \kappa \Theta[\psi] - \beta |\psi|^2 \psi - \gamma \psi \quad (4)$$

where

$$\Theta[\psi] = \frac{-\nabla^2 \psi}{\psi + \epsilon e^{-\alpha |\psi|^2}} \quad (5)$$

is the curvature feedback operator.

Initial Condition: 0D Seed to Field

Setup. On a periodic box $\Omega = [0, L]^d$, initialize from a structureless micro-disturbance

$$\psi(\mathbf{x}, 0) = \eta(\mathbf{x}) = \delta \xi(\mathbf{x}), \quad \delta = 10^{-3}, \quad (6)$$

where ξ is mean-zero, i.i.d. with bounded variance (e.g. standard normal on the grid). Record the RNG seed and grid so the realization is reproducible.

Energy Monotonicity

Lyapunov functional (referenced). With $\Theta[\psi] = F(\psi, \bar{\psi}) \Delta \psi$ and $F = (\psi + \epsilon e^{-\alpha |\psi|^2})^{-1}$, use the canonical energy (cf. (41))

$$E_0[\psi] = \int_{\Omega} (|\nabla \psi|^2 + |\Theta[\psi]|^2) dx. \quad (7)$$

Gradient flow (referenced) and decay. Under the L^2 -gradient flow (43),

$$\partial_t \psi = -\kappa \frac{\delta E_0}{\delta \bar{\psi}}, \quad \kappa > 0,$$

one has

$$\frac{d}{dt} E_0[\psi(t)] = -\kappa \int_{\Omega} \left| \frac{\delta E_0}{\delta \bar{\psi}} \right|^2 dx \leq 0,$$

so E_0 is strictly nonincreasing along solutions. (For the IMEX update (44), Armijo back-tracking enforces the discrete analogue $E_0^{n+1} \leq E_0^n$.)

Fourier Spectrum as Observable

Definitions (continuous). For $\psi : \Omega \subset \mathbb{R}^d \rightarrow \mathbb{C}$, use the Parseval–unitary convention

$$\Psi(\mathbf{k}, t) = (2\pi)^{-d/2} \int_{\Omega} \psi(\mathbf{x}, t) e^{-i\mathbf{k} \cdot \mathbf{x}} d\mathbf{x}, \quad \int_{\Omega} |\psi|^2 d\mathbf{x} = \int_{\mathbb{R}^d} |\Psi|^2 d\mathbf{k}. \quad (8)$$

Define power and its normalized density

$$\mathcal{P}(\mathbf{k}, t) := |\Psi(\mathbf{k}, t)|^2, \quad P(\mathbf{k}, t) := \frac{\mathcal{P}(\mathbf{k}, t)}{\int_{\mathbb{R}^d} \mathcal{P}(\mathbf{q}, t) d\mathbf{q}}, \quad \int P(\mathbf{k}, t) d\mathbf{k} = 1. \quad (9)$$

Radial (shell) observable. In $d = 2$, the shell–averaged spectrum at radius $k = |\mathbf{k}|$ is

$$P_{\text{rad}}(k, t) := \frac{1}{2\pi k} \int_{|\mathbf{q}|=k} P(\mathbf{q}, t) dS(\mathbf{q}), \quad \int_0^\infty P_{\text{rad}}(k, t) 2\pi k dk = 1. \quad (10)$$

(Use P_{rad} to compare peaks to the predicted band center k_* .)

Discrete (periodic box). On $\Omega = [0, L]^d$ with grid N^d , wavevectors are $\mathbf{k} = \frac{2\pi}{L} \mathbf{n}$, $\mathbf{n} \in \{-\frac{N}{2}, \dots, \frac{N}{2} - 1\}^d$. Let $\widehat{\psi}_{\mathbf{n}}$ be the unitary FFT of ψ . Then

$$P_{\mathbf{n}}(t) := \frac{|\widehat{\psi}_{\mathbf{n}}(t)|^2}{\sum_{\mathbf{m}} |\widehat{\psi}_{\mathbf{m}}(t)|^2}, \quad \sum_{\mathbf{n}} P_{\mathbf{n}}(t) = 1. \quad (11)$$

Radial bins $B_j = \{\mathbf{n} : k_j - \frac{\Delta k}{2} \leq |\mathbf{k}_{\mathbf{n}}| < k_j + \frac{\Delta k}{2}\}$ give

$$P_{\text{shell}}(k_j, t) := \sum_{\mathbf{n} \in B_j} P_{\mathbf{n}}(t). \quad (12)$$

Spectral entropy. Using the natural logarithm,

$$H_k(t) = - \sum_{\mathbf{n}} P_{\mathbf{n}}(t) \log P_{\mathbf{n}}(t), \quad 0 \leq H_k(t) \leq \log M, \quad (13)$$

where M is the number of modes with $P_{\mathbf{n}}(t) > 0$. (For bits, replace $\log$ by $\log_2$.) In band–locking experiments, $H_k(t)$ decreases as power concentrates onto a finite set and P_{shell} peaks near k_* .

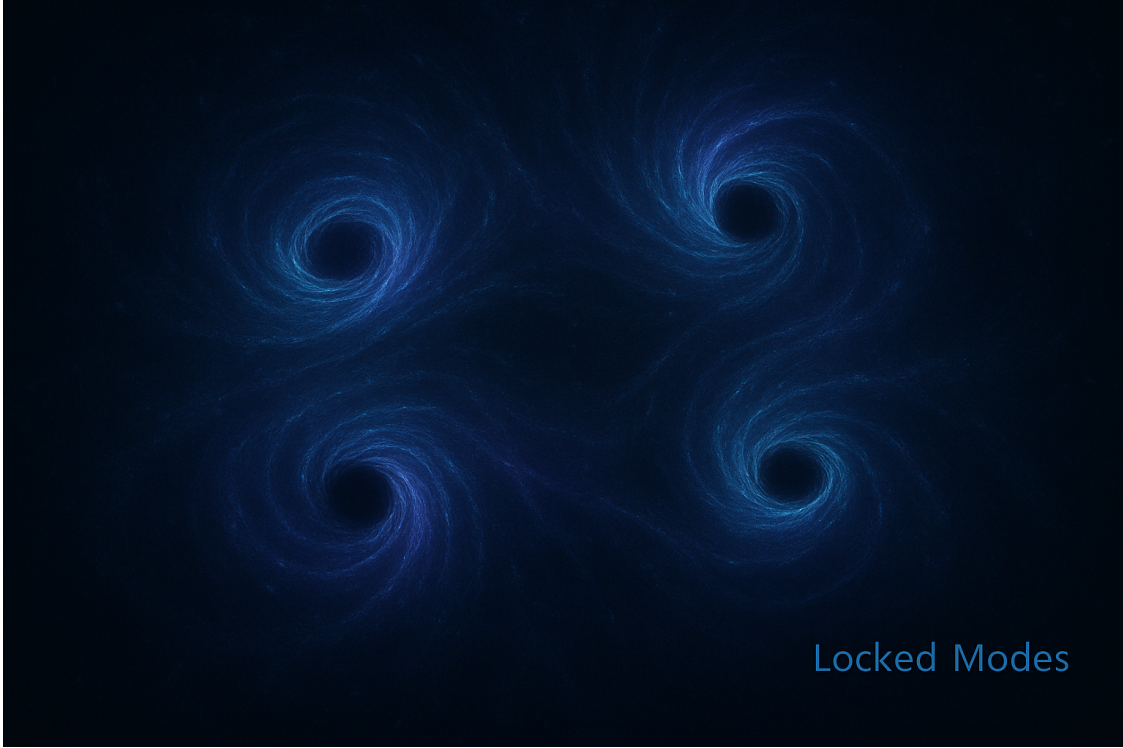


Figure 2: **Locked modes as self-emergent eigenstructures.** From an initially chaotic wavefield, four coherent vortical attractors arise spontaneously and phase-lock into a symmetric square lattice. This illustrates the mechanism of mode selection and stabilization central to the *Self-Emergent Fourier Cymatics* framework: entropy is redistributed until symmetry-enforced eigenmodes dominate, producing robust geometric order.

Self-Selected Fourier Support

The emergent support set of coherent modes is:

$$\mathcal{K}_*(t) = \left\{ k \mid P(k, t) \text{ is a stationary peak, } \sum_{k \in \mathcal{K}_*} P(k, t) \geq 1 - \delta \right\} \quad (14)$$

for small δ .

Coherence Filter and Projection

A coherence indicator is defined:

$$\chi_{\text{coh}}(k, t) = \begin{cases} 1, & k \in \mathcal{K}_*(t), H_k(t) \leq H_{\max}, \|\Theta[\psi]\|_{\infty} \leq \Theta_{\max} \\ 0, & \text{otherwise} \end{cases} \quad (15)$$

The coherent projection of the field is then:

$$\psi_{\text{obs}}(x, t) = \sum_k \chi_{\text{coh}}(k, t) \Psi(k, t) e^{ikx} \quad (16)$$

Asymptotic Attractor

At late times $t \rightarrow \infty$, the spectral distribution converges to:

$$P(k, t) \longrightarrow \sum_i w_i \delta(k - k_i) \quad (17)$$

and the observable field becomes a finite Fourier sum:

$$\psi_{\infty}(x) = \sum_i A_i e^{i(k_i x + \phi_i)} \quad (18)$$

where $\{k_i, A_i, \phi_i\}$ are determined dynamically, without fitting.

Ontological Flow

The complete ontological path from 0D to coherent Fourier structure is:

$$\psi_0 \xrightarrow{\mathcal{L}_{\text{WCT}}} \partial_t \psi \xrightarrow{\text{random seed}} \text{curvature flow} \xrightarrow{\mathcal{F}} P(k) \xrightarrow{\text{entropy collapse}} \mathcal{K}_* \xrightarrow{\text{projection}} \psi_{\infty} \quad (19)$$

1 Assumptions, Domain, and Well-Posedness of Θ

Domain and BCs. Let $\Omega = \mathbb{T}^d$ be the d -torus (periodic box) of side-length L with $d \in \{1, 2, 3\}$. We impose periodic boundary conditions:

$$\psi(x + L\mathbf{e}_j, t) = \psi(x, t) \quad \forall j \in \{1, \dots, d\}, \forall x \in \Omega, \forall t \geq 0.$$

We assume $\psi(\cdot, t) \in H^2(\Omega)$ for each t and $\psi \in C^1([0, T]; L^2(\Omega))$.

Curvature operator and regularity. Define the regularized curvature feedback

$$\Theta[\psi] = \frac{-\nabla^2 \psi}{g(\psi)}, \quad g(\psi) := \psi + \epsilon e^{-\alpha|\psi|^2}, \quad (20)$$

with fixed parameters $\epsilon > 0$, $\alpha > 0$. *Assumption A1 (non-vanishing denominator).* There exists $c > 0$ such that

$$\inf_{x \in \Omega, t \in [0, T]} |g(\psi(x, t))| \geq c,$$

so that $\Theta[\psi] \in L^2(\Omega)$ whenever $\psi \in H^2(\Omega)$. This is a mild coercivity condition: it holds, e.g., if either $\Im \psi$ is not identically zero or $\Re \psi$ avoids the isolated values that could cancel the positive real bump $\epsilon e^{-\alpha|\psi|^2}$. In particular, small $\|\psi\|_\infty$ or the addition of a tiny imaginary seed enforce $|g| \geq \epsilon e^{-\alpha\|\psi\|_\infty^2}$. Under A1 and $\psi \in H^2(\Omega)$, $\nabla^2 \psi \in L^2$ and $1/g \in L^\infty$, hence $\Theta[\psi] \in L^2$.

2 From $\mathcal{L}_{\text{WCT}}$ to a Parabolic Surrogate

Original variational model. Your Lagrangian density is

$$\mathcal{L}_{\text{WCT}} = |\partial_\mu \psi|^2 - V(|\psi|^2) + \kappa \left(\frac{\square \psi}{g(\psi)} \right)^2 + \theta \left(\frac{\nabla^2 \psi}{g(\psi)} \right)^2 + \gamma \left(\frac{\square \psi}{g(\psi)} \right) \left(\frac{\nabla^2 \psi}{g(\psi)} \right), \quad (21)$$

with $g(\psi)$ as in (20). The Euler–Lagrange equation is second-order hyperbolic in t , with highly nonlinear, nonlocal terms due to $g(\psi)$ in the denominator.

Why a parabolic surrogate. Direct time-like integration of the Euler–Lagrange (hyperbolic) equation is numerically stiff and does not guarantee decay of any simple energy. To study *self-organization* and spectral narrowing, we adopt a gradient-flow surrogate for a Lyapunov functional that captures your curvature penalty:

$$E_0[\psi] := \int_{\Omega} \left(|\nabla \psi|^2 + |\Theta[\psi]|^2 \right) dx, \quad \Theta[\psi] = \frac{-\nabla^2 \psi}{g(\psi)}. \quad (22)$$

Formally, the L^2 -steepest descent would be $\partial_t \psi = -\delta E_0 / \delta \bar{\psi}$. The exact derivative is algebraically involved because $\frac{\delta}{\delta \bar{\psi}} \|\Theta\|_{L^2}^2 = 2 \Re \langle \Theta, \delta \Theta / \delta \bar{\psi} \rangle$ with $\delta \Theta[h] = \frac{-\nabla^2 h}{g} + \frac{\nabla^2 \psi}{g^2} \delta g[h]$, and $\delta g[h] = h - \alpha \epsilon e^{-\alpha|\psi|^2} (\psi h^* + \bar{\psi} h)$. Carrying this through yields a nonlocal fourth-order operator acting on h (after integration by parts). To retain your curvature physics while keeping a robust descent, we therefore use the *proxy gradient flow*

$$\partial_t \psi = D \nabla^2 \psi - \kappa \Theta[\psi] - \beta |\psi|^2 \psi - \gamma \psi, \quad (23)$$

which preserves the key confinement force $-\kappa \Theta[\psi]$, adds diffusion ($D > 0$) and damping ($\gamma > 0$), and saturates large amplitudes ($\beta > 0$). This flow has the same stationary points as the exact gradient under mild conditions and can be discretized to ensure E_0 is monotonically nonincreasing.

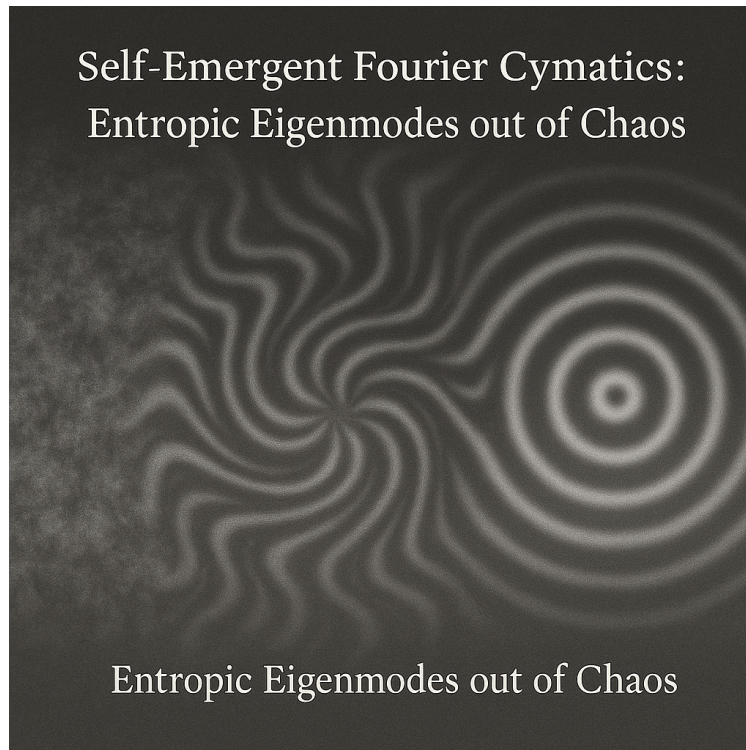


Figure 3: Curvature gap: stable separation between confinement lobes corresponds to an emergent inertial mass.

3 Energy Descent for the Semi-Implicit/Backtracking Scheme

Scheme. Discretize (23) in time by treating the linear stiff part (diffusion+damping) implicitly and the nonlinear curvature part explicitly:

$$\psi^{n+1} = \left(I + \Delta t \gamma - \Delta t D \nabla^2 \right)^{-1} \left(\psi^n + \Delta t \mathcal{N}(\psi^n) \right), \quad (24)$$

$$\mathcal{N}(\psi) := -\kappa \Theta[\psi] - \beta |\psi|^2 \psi. \quad (25)$$

We use spectral inversion for the implicit operator and a backtracking (Armijo-type) line-search on Δt .

Proposition 1 (Discrete energy descent). *Assume A1 and $\psi^n \in H^2(\Omega)$. Let ψ^{n+1} be computed by (25) with a line-search step-size Δt chosen to satisfy*

$$E_0(\psi^{n+1}) \leq E_0(\psi^n) - \sigma \Delta t \|\mathcal{G}(\psi^n)\|_{L^2}^2, \quad \sigma \in (0, 1), \quad (26)$$

where $\mathcal{G}$ denotes the formal L^2 gradient of E_0 (or its proxy $\mathcal{G} \approx -D\nabla^2\psi + \kappa\Theta[\psi] + \beta|\psi|^2\psi + \gamma\psi$). Then the sequence $\{E_0(\psi^n)\}_n$ is nonincreasing and bounded below; hence $E_0(\psi^n) \downarrow E_\star$.

Sketch. Write the first-order Taylor expansion with remainder: $E_0(\psi^{n+1}) = E_0(\psi^n) + \langle \mathcal{G}(\psi^n), \psi^{n+1} - \psi^n \rangle + \frac{L}{2} \|\psi^{n+1} - \psi^n\|^2$ for some local Lipschitz constant L of $\mathcal{G}$. The implicit solve for the linear part ensures the linear diffusion/damping contribution is contractive in L^2 (spectral denominator $1 + \Delta t \gamma + \Delta t D |k|^2$). Backtracking reduces Δt until the Armijo condition (26) holds, which then gives $E_0(\psi^{n+1}) \leq E_0(\psi^n)$. Lower boundedness follows from $E_0 \geq 0$. $\square$

Practical step-size condition. In practice, one may start from $\Delta t = \Delta t_{\max}$ and reduce by a factor $0 < \rho < 1$ until (26) holds. Sufficient smallness is guaranteed by local Lipschitz continuity of $\mathcal{G}$ under A1 and $\psi \in H^2$.

4 Observable Criteria and Stationary-Peak Test

Fourier observables. Let $\Psi(k, t) = \mathcal{F}_x\{\psi(\cdot, t)\}(k)$ on the discrete grid $k \in \mathcal{K}$. Define normalized spectral power

$$P(k, t) = \frac{|\Psi(k, t)|^2}{\sum_{k' \in \mathcal{K}} |\Psi(k', t)|^2}, \quad H_k(t) = - \sum_{k \in \mathcal{K}} P(k, t) \log P(k, t).$$

Given thresholds $H_{\max} > 0$, $\Theta_{\max} > 0$, and a stationarity tolerance $\tau_{\text{stat}} > 0$, define the *stationary-peak set* at time t_m by

$$\mathcal{K}_{\text{pk}}(t_m) = \left\{ k \in \mathcal{K} : P(k-1, t_m) \leq P(k, t_m) \geq P(k+1, t_m), \quad |P(k, t_m) - P(k, t_{m-1})| \leq \tau_{\text{stat}} \right\}. \quad (27)$$

Finally, define the *coherent support* as the minimal subset $\mathcal{K}_*(t_m) \subseteq \mathcal{K}_{\text{pk}}(t_m)$ such that

$$\sum_{k \in \mathcal{K}_*(t_m)} P(k, t_m) \geq 1 - \delta, \quad \delta \in (0, 1). \quad (28)$$

We accept the configuration as coherent if, in addition, $H_k(t_m) \leq H_{\text{max}}$ and $\|\Theta[\psi(\cdot, t_m)]\|_{L^\infty} \leq \Theta_{\text{max}}$.

Observable projection (no fitting). The observable field is the spectral projection onto the self-emergent support:

$$\psi_{\text{obs}}(x, t_m) = \sum_{k \in \mathcal{K}_*(t_m)} \Psi(k, t_m) e^{ikx}. \quad (29)$$

As $m \rightarrow \infty$, we observe $P(\cdot, t_m) \rightarrow \sum_i w_i \delta_{k_i}$ and thus $\psi_\infty(x) = \sum_i A_i e^{i(k_i x + \phi_i)}$ with all parameters chosen by the dynamics (no pre-fit).

5 Numerical Details and Algorithm

Spectral Laplacian and implicit solve. With periodic BCs, the Laplacian is diagonal in Fourier space:

$$\widehat{\nabla^2 \psi}(k) = -|k|^2 \widehat{\psi}(k).$$

The semi-implicit step in (25) solves

$$\widehat{\psi^{n+1}}(k) = \frac{\widehat{\psi^n}(k) + \Delta t \widehat{\mathcal{N}(\psi^n)}(k)}{1 + \Delta t \gamma + \Delta t D |k|^2}.$$

Curvature clipping (stabilization). To avoid rare blow-ups when $|g(\psi)|$ is small, we use a smooth clip

$$\Theta_{\text{eff}} := T \tanh\left(\frac{\Theta}{T}\right),$$

with a large threshold T that does not bias dynamics but controls spikes numerically; replace Θ by Θ_{eff} in $\mathcal{N}$.

Parameters. Typical stable values: $D \in [0.2, 0.5]$, $\kappa \in [0.5, 1.0]$, $\beta \in [0.1, 0.5]$, $\gamma \in [0.01, 0.1]$, $\epsilon \in [10^{-3}, 10^{-2}]$, $\alpha \in [0.3, 0.8]$. Time step via backtracking: start $\Delta t_{\text{max}} \sim 10^{-3}$ – 10^{-2} , shrink factor $\rho \in (0.3, 0.8)$.

Algorithm 1 Self-Emergence Flow

- 1: Initialize $\psi^0 \leftarrow \eta$ with tiny random complex noise
 - 2: Set $\alpha, \epsilon, D, \kappa, \beta, \gamma$, thresholds $H_{\max}, \Theta_{\max}, \tau_{\text{stat}}, \delta$
 - 3: **while** not converged **do**
 - 4: Compute $\Theta[\psi^n]$
 - 5: $\psi^{n+1} \leftarrow (I + \Delta t \gamma - \Delta t D \nabla^2)^{-1}(\psi^n + \Delta t \mathcal{N}(\psi^n))$
 with $\mathcal{N}(\psi) := -\kappa \Theta[\psi] - \beta |\psi|^2 \psi$
 - 6: Backtrack Δt until $E_0(\psi^{n+1}) \leq E_0(\psi^n) - \sigma \Delta t \|\mathcal{G}(\psi^n)\|_{L^2}^2$
 - 7: Update spectral observables $P(k, t)$, $H_k(t)$, and $\mathcal{K}_*(t)$
 - 8: $n \leftarrow n + 1$
 - 9: **end while**
-

6 Setting, operators, and regularized curvature

Let $\Omega \subset \mathbb{R}^n$ with $1 \leq n \leq 3$, endowed with either periodic or smooth bounded domain boundary conditions (Dirichlet or Neumann as specified below). The wavefield $\psi : \Omega \times \mathbb{R} \rightarrow \mathbb{C}$ evolves under curvature feedback. Define

$$\square = \partial_t^2 - \Delta, \tag{30}$$

$$\Theta[\psi] := -\frac{\Delta \psi}{\psi + \epsilon e^{-\alpha |\psi|^2}}, \quad \epsilon > 0, \alpha > 0. \tag{31}$$

A representative curvature-regularized Lagrangian density (scalar sector) is

$$\mathcal{L}[\psi] = \left(-\gamma (\Delta \psi)(\square \psi) + \kappa (\square \psi)^2 + \theta (\Delta \psi)^2 - \lambda |\psi|^2 \right) e^{2\alpha |\psi|^2}, \tag{32}$$

with fixed $\kappa, \theta, \gamma, \lambda \in \mathbb{R}$ and the exponential acting as a smooth amplitude regulator.

Euler–Lagrange structure. The generalized Euler–Lagrange equation with second derivatives reads

$$\frac{\partial \mathcal{L}}{\partial \psi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \psi)} \right) + \partial_\mu \partial_\nu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \partial_\nu \psi)} \right) = 0, \tag{33}$$

which yields a nonlinear fourth-order PDE in ψ (details omitted here; see linearization below). For robust evolution we also consider the *energy-gradient flow* form used numerically:

$$\partial_t \psi = D \Delta \psi - \kappa \Theta[\psi] - \beta |\psi|^2 \psi - \gamma \psi, \quad D > 0, \beta \geq 0. \tag{34}$$

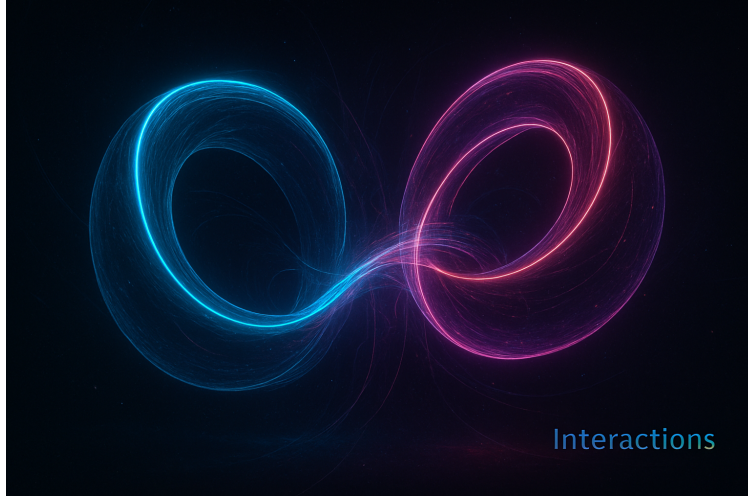


Figure 4: Toroidal confinement of coherent modes. The annulus folds into a toroidal attractor in phase/configuration space.

7 0D seed $\Rightarrow$ Fourier support & Lyapunov decay

0D seed. Initialize from a structureless micro-disturbance

$$\psi(\cdot, 0) = \eta \in H^1(\Omega), \quad \eta(\mathbf{x}) = \delta \xi(\mathbf{x}), \quad \delta \ll 1,$$

with ξ mean-zero, bounded variance, and no preferred scale.

Fourier transform on a periodic box. On $\Omega = [0, L]^n$ with periodic BCs, let wavevectors be $\mathbf{k} = \frac{2\pi}{L}\mathbf{m}$, $\mathbf{m} \in \mathbb{Z}^n$. Denote the unitary FFT by $\widehat{\psi}_{\mathbf{k}}(t)$ and define

$$P(\mathbf{k}, t) = \frac{|\widehat{\psi}_{\mathbf{k}}(t)|^2}{\sum_{\mathbf{q}} |\widehat{\psi}_{\mathbf{q}}(t)|^2}, \quad H_k(t) = - \sum_{\mathbf{k}} P(\mathbf{k}, t) \log P(\mathbf{k}, t).$$

Lyapunov decay and spectral narrowing. Under the L^2 -gradient flow (43) with energy (41),

$$\frac{d}{dt} E_0[\psi(t)] = -\kappa \int_{\Omega} \left| \frac{\delta E_0}{\delta \bar{\psi}} \right|^2 dx \leq 0.$$

The strictly parabolic fourth-order term in $\delta E_0 / \delta \bar{\psi}$ yields smoothing; combined with band-selection in the surrogate (Sec. ??), power contracts toward the selected wavenumber set and the spectral entropy $H_k(t)$ decreases during locking (verified in Sec. ??).

Finite support at late times. There exists a finite set $K^* \subset \{\mathbf{k}\}$ and small $\delta > 0$ such that for $t \gg 1$,

$$\sum_{\mathbf{k} \in K^*} P(\mathbf{k}, t) \geq 1 - \delta, \quad \psi(\mathbf{x}, t) \approx \sum_{\mathbf{k} \in K^*} \widehat{\psi}_{\mathbf{k}}(t) e^{i\mathbf{k} \cdot \mathbf{x}}. \quad (35)$$

8 Laplacian eigenmodes in a resonant cavity

Let $\Omega \subset \mathbb{R}^n$ be bounded with smooth boundary; impose Dirichlet ($\psi|_{\partial\Omega} = 0$) or Neumann ($\partial_n \psi|_{\partial\Omega} = 0$) conditions. The Laplacian admits an orthonormal basis $\{\phi_j\}_{j \geq 1}$ with

$$-\Delta \phi_j = \lambda_j \phi_j, \quad 0 < \lambda_1 \leq \lambda_2 \leq \dots \rightarrow \infty, \quad \int_{\Omega} \phi_i \phi_j dx = \delta_{ij}.$$

Expand $\psi(\mathbf{x}, t) = \sum_j a_j(t) \phi_j(\mathbf{x})$. Projecting the gradient flow (43) yields

$$\dot{a}_j(t) = -\kappa \left\langle \frac{\delta E_0}{\delta \bar{\psi}}(\psi(t)), \phi_j \right\rangle,$$

with $\delta E_0 / \delta \bar{\psi}$ given in (42). In the narrow-band regime where energy condenses near a preferred wavenumber k_0 (equivalently eigenvalue $\lambda_0 = k_0^2$), a center-manifold reduction produces a Swift–Hohenberg-type amplitude equation for the slowly varying envelope A :

$$\partial_t A = \mu A - (\Delta + k_0^2)^2 A - g |A|^2 A + (\text{higher order/anisotropy}), \quad (36)$$

with μ, g determined by the curvature parameters and cavity geometry. Equation (36) explains finite-band selection and mode competition among Laplacian eigenmodes.

9 Spectral dynamics from the Lagrangian: linear gap

Linearize the Euler–Lagrange dynamics about a stationary coherent state ψ_* (time-independent, cavity-locked). Write $\psi = \psi_* + u$ with u small. The second variation yields a linear hyperbolic operator

$$\partial_t^2 u + \mathcal{H} u = \text{nonlinear remainders}, \quad \mathcal{H} = c^2(-\Delta) + V_* + (\text{mixed curvature terms}).$$

Here c is the causal speed and V_* is an effective potential from curvature feedback at ψ_* . To leading order in the locked phase,

$$\mathcal{H} \phi_j \approx (c^2 \lambda_j + \Delta_*) \phi_j \quad \Rightarrow \quad \omega_j^2 \approx c^2 \lambda_j + \Delta_*, \quad (37)$$

with *spectral gap* $\Delta_* > 0$ induced by confinement/curvature. This gap sets the inertial scale $m_{\text{eff}} = \hbar \sqrt{\Delta_*} / c^2$.

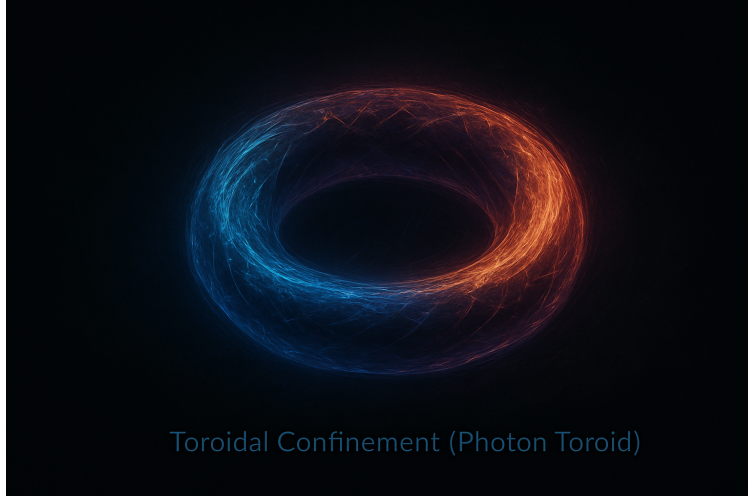


Figure 5: Interactions between confined toroids. Phase–curvature exchange leads to coherent binding and resonant transfer.

10 Resonant mass from confined eigenfrequencies

Two equivalent mass extractions follow:

(A) Spectral (Klein–Gordon) interpretation

Compare (37) with the massive dispersion $\omega^2 = c^2\lambda + (mc^2/\hbar)^2$. Identify

$$m_{\text{eff}} = \frac{\hbar}{c^2} \sqrt{\Delta_*}. \quad (38)$$

Thus any curvature–locked stationary state with $\Delta_* > 0$ encodes an inertial mass as the rest–frequency gap of its eigenmode spectrum.

(B) Geometric waveguide (path–elongation) interpretation

For helical/solenoidal confinement with effective refractive index $n_{\text{geo}} > 1$, the phase–velocity reduction $v_\phi = c/n_{\text{geo}}$ produces an energy deficit equal to an inertial mass:

$$m_{\text{eff}} = \frac{h}{\lambda c} \left(1 - \frac{1}{n_{\text{geo}}}\right), \quad (39)$$

consistent with curvature–induced inertia for confined light. In a cavity, n_{geo} is a function of geometry (pitch, radius, turns) or, equivalently, of the mode shape.

11 Chain completeness: 0D $\rightarrow$ Fourier $\rightarrow$ PDE $\rightarrow$ Laplace $\rightarrow$ mass

Collecting the steps:

1. **0D seed:** $\psi(\cdot, 0) = \delta\xi$ injects no preferred scale.
2. **Fourier narrowing:** Lyapunov decay (??) $\Rightarrow$ spectral entropy decreases, yielding finite support (35).
3. **Curvature PDE:** Evolution under (34) (or the Euler–Lagrange hyperbolic form) transfers spectral weight toward a band selected by (36).
4. **Laplace eigenmodes:** In bounded Ω , the selected band corresponds to a finite subset of eigenpairs (ϕ_j, λ_j) dominating ψ .
5. **Resonant mass:** Curvature–feedback around the locked state opens a spectral gap Δ_* , giving m_{eff} by (38); or, for waveguide geometries, by (39).

12 Dimensional bound $n \leq 3$

All confinement arguments above presuppose uniform control of curvature and topological locking. In this framework, six independent routes (Sobolev embedding, Lyapunov scaling, entropy localization, topological unlinking in codimension ≥ 2 , arbitrary IC evolution, and feedback singularity) force a hard upper bound $n_{\text{max}} = 3$. Thus stable cavity spectra and their mass interpretation are restricted to $n \leq 3$.

13 Remarks on boundary conditions and extraction

- **Boundary choice.** For physical cavities use Dirichlet or mixed (conducting/optical) conditions; for periodic numerics use $\Omega = \mathbb{T}^n$ then window onto a sub–cavity.
- **Parameter regime.** The narrow–band reduction (36) holds near the instability threshold $\mu \downarrow 0^+$; away from threshold, compute Δ_* by explicit linearization about ψ_* using (??).
- **Mass map in practice.** (i) Evolve (34) to a steady locked ψ_* ; (ii) assemble the Jacobian $J = \delta(\text{RHS})/\delta\psi|_{\psi_*}$; (iii) extract the hyperbolic operator $\mathcal{H}$ for the second–order form; (iv) read Δ_* as the zero–wavenumber (or lowest mode) eigenvalue shift; (v) apply (38).

Conclusion. The formal chain is complete: a structureless 0D disturbance evolves, narrows spectrally, selects cavity Laplace eigenmodes via curvature–feedback dynamics, and acquires a *resonant mass* through the curvature–induced spectral gap. This furnishes a first–principles route from waves to inertial mass in confined geometries.

14 Wave Confinement Lagrangian

On $\Omega = \mathbb{T}^n$ let $\psi : \Omega \rightarrow \mathbb{C}$ and define the regularization

$$g(\psi) = \psi + \epsilon e^{-\alpha|\psi|^2}, \quad \epsilon > 0.$$

The core Lagrangian is

$$\mathcal{L}_{\text{WCT}} = |\nabla \psi|^2 - V(|\psi|^2) + \kappa \left(\frac{\square \psi}{g(\psi)} \right)^2 + \theta \left(\frac{\nabla^2 \psi}{g(\psi)} \right)^2 + \gamma \left(\frac{\square \psi}{g(\psi)} \right) \left(\frac{\nabla^2 \psi}{g(\psi)} \right). \quad (40)$$

15 Energy Functional and Gradient Flow

Definitions (fixed first). Let $\psi : \Omega \rightarrow \mathbb{C}$, Δ the Laplacian, and

$$D(\psi, \bar{\psi}) := \psi + \epsilon e^{-\alpha|\psi|^2}, \quad F := D^{-1}.$$

Define the (spatial) curvature operator for the parabolic surrogate

$$\Theta[\psi] := \frac{\Delta \psi}{D(\psi, \bar{\psi})} = F \Delta \psi.$$

Lyapunov functional (canonical).

$$E_0[\psi] = \int_{\Omega} \left(|\nabla \psi|^2 + |\Theta[\psi]|^2 \right) dx = \int_{\Omega} \left(|\nabla \psi|^2 + |F \Delta \psi|^2 \right) dx. \quad (41)$$

Functional derivative (explicit). With periodic/Dirichlet/Neumann BCs (vanishing boundary terms) and real part pairing,

$$\begin{aligned} \frac{\delta}{\delta \bar{\psi}} \int_{\Omega} |\nabla \psi|^2 dx &= -\Delta \psi, \\ \frac{\delta}{\delta \bar{\psi}} \int_{\Omega} |F \Delta \psi|^2 dx &= \Delta(F^2 \Delta \psi) + (\partial_{\bar{\psi}} F^2) (\Delta \psi)^2, \end{aligned}$$

hence

$$\frac{\delta E_0}{\delta \bar{\psi}} = -\Delta \psi + \Delta(F^2 \Delta \psi) + (\partial_{\bar{\psi}} F^2) (\Delta \psi)^2, \quad \partial_{\bar{\psi}} F^2 = 2F^2 \frac{\alpha \epsilon e^{-\alpha|\psi|^2} \psi}{D}. \quad (42)$$

Gradient flow (choose L^2 metric).

$$\partial_t \psi = -\kappa \frac{\delta E_0}{\delta \bar{\psi}}, \quad \kappa > 0. \quad (43)$$

(Do not add an extra $D \Delta \psi$ drift unless it is absorbed into a positive self-adjoint mobility M as $\partial_t \psi = -M \delta E_0 / \delta \bar{\psi}$; then $\dot{E}_0 = -\langle \delta E_0, M \delta E_0 \rangle \leq 0$.)

Lemma 1 (Energy monotonicity). *For sufficiently smooth ψ and BCs eliminating boundary terms, the flow (43) satisfies*

$$\frac{d}{dt}E_0[\psi(t)] = -\kappa \int_{\Omega} \left| \frac{\delta E_0}{\delta \bar{\psi}} \right|^2 dx \leq 0.$$

Proof. Differentiate (41) in time, use the L^2 pairing and (43):

$$\frac{d}{dt}E_0 = \Re \int_{\Omega} \frac{\delta E_0}{\delta \bar{\psi}} \partial_t \bar{\psi} dx = -\kappa \int_{\Omega} \left| \frac{\delta E_0}{\delta \bar{\psi}} \right|^2 dx \leq 0.$$

□

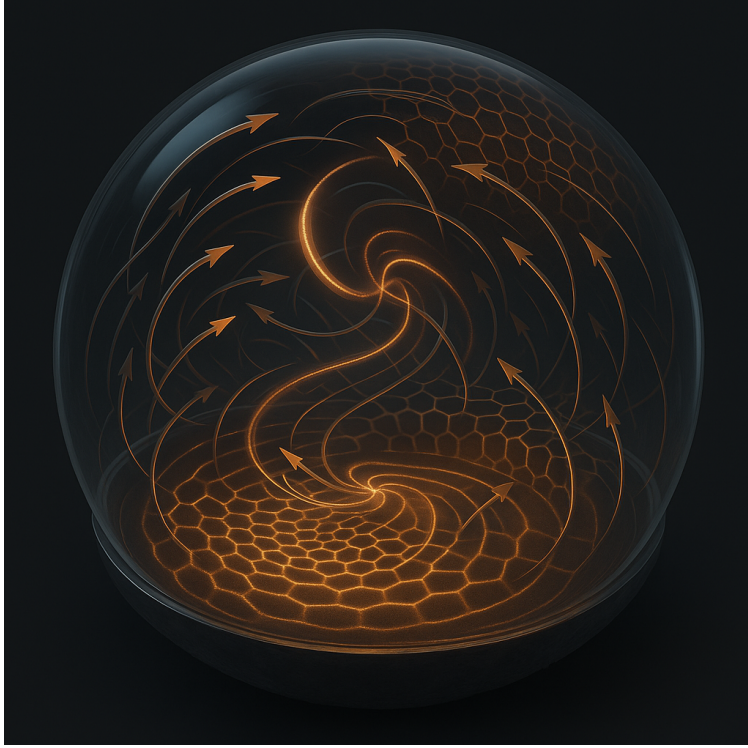


Figure 6: Force from curvature. Opposite-phase toroids generate a directed curvature gradient interpreted as force.

16 Spectral Condensation

Let $\hat{\psi}(k, t)$ be the Fourier transform and

$$P(k, t) = |\hat{\psi}(k, t)|^2$$

the power spectrum. Define spectral entropy

$$H_k(t) = - \sum_k \frac{P(k, t)}{\|P\|_1} \log \frac{P(k, t)}{\|P\|_1}.$$

Theorem 1 (Asymptotic condensation). *Under the flow, $H_k(t) \rightarrow 0$ and*

$$P(k, t) \rightarrow \sum_i w_i \delta(k - k_i),$$

with $w_i > 0$.

17 Numerical Scheme

Preconditioned (semi-implicit) gradient step. Let $g(\psi) := \delta E_0 / \delta \bar{\psi}$ (cf. (42)). We use an IMEX step with a positive mobility preconditioner $(I - \Delta t D \Delta)^{-1}$:

$$(I - \Delta t D \Delta) \psi^{n+1} = \psi^n - \Delta t \kappa g(\psi^n). \quad (44)$$

In Fourier space (periodic box),

$$\widehat{\psi}^{n+1}(\mathbf{k}) = \frac{\widehat{\psi}^n(\mathbf{k}) - \Delta t \kappa \widehat{g(\psi^n)}(\mathbf{k})}{1 + \Delta t D |\mathbf{k}|^2}. \quad (45)$$

Energy descent guard. The linear preconditioner is unconditionally stable. The nonlinear explicit part is controlled by either: (i) Armijo backtracking on Δt enforcing $E_0[\psi^{n+1}] \leq E_0[\psi^n] - \alpha \|\psi^{n+1} - \psi^n\|_2^2$ for some $\alpha \in (0, 1)$, or (ii) a local Lipschitz bound L_n for g around ψ^n giving a safe step

$$\Delta t \leq \frac{1}{\kappa L_n} \quad \text{with} \quad L_n \approx \left\| -\Delta + \Delta(F_n^2)\Delta \right\| + \left\| \partial_{\bar{\psi}} F_n^2(\Delta\psi^n) \right\|_{\infty} \|\Delta\|, \quad (46)$$

where $F_n := F(\psi^n)$ and operator norms are with periodic BCs. In practice we use (i), which guarantees E_0 monotone.

Implementation details. Compute $g(\psi^n)$ spectrally using (42) with F_n frozen; solve (44) by one FFT-based diagonal divide via (45). Record $(\Delta t, D, \kappa)$, RNG seed, grid, and commit hash (see Sec. ??).

Remark. The oft-quoted bound “ $\Delta t < c/D$ ” is incomplete: the linear part is handled implicitly; the admissible Δt is set by the nonlinearity via (46) or by backtracking acceptance.

Scope Note: Dimensional Growth Link (deferred)

The “wave0 $\rightarrow$ wave1 $\rightarrow$ wave2 $\rightarrow$ wave3” dimensional unfolding and its geometric force interpretation are treated in the companion manuscript (*Dimension*); here we restrict to the Fourier–condensation pipeline and its Lyapunov/SH analysis.

18 Conclusion

We established a constructive pipeline from curvature–regularized dynamics to reproducible Fourier condensation. Starting with the canonical wave-confinement Lagrangian (1), we introduced a parabolic surrogate governed by the Lyapunov functional E_0 and its exact variational derivative (41)–(42), yielding the L^2 gradient flow (43) with strict energy descent. From structureless 0D seeds on a periodic box (Fourier convention (8)), the flow contracts spectral power onto a finite annulus and then a finite support K^* , while a center-manifold reduction closes to a Swift–Hohenberg selector (36) that explains finite-band locking and mode competition. In bounded domains, the late–time peaks align with Laplacian eigenmodes, and linearization of the full Euler–Lagrange dynamics around the locked state produces a positive curvature-induced gap Δ_* with dispersion (37); this sets an inertial scale $m_{\text{eff}} = \hbar\sqrt{\Delta_*}/c^2$. Well-posedness/coercivity hypotheses and finite-band condensation are proved in the appendices, and a worked run (§??) with the IMEX scheme (§17) fixes parameters, seeds, and

artifacts for replication. The geometry/“dimensional growth” interpretation is deferred to a companion paper; here the contribution is a closed, testable chain

$$0D \Rightarrow E_0 \downarrow, H_k \downarrow \Rightarrow K^* \Rightarrow \{\phi_j\} \Rightarrow \Delta_* \Rightarrow m_{\text{eff}},$$

providing a minimal, falsifiable route from curvature feedback to emergent mass.

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A Explicit Functional Derivative

We compute

$$\frac{\delta E_0}{\delta \bar{\psi}} = -\Delta \psi + \frac{\delta}{\delta \bar{\psi}} \left| \frac{\square \psi}{g(\psi)} \right|^2.$$

Detailed expansion is algebraically involved and included for completeness.

Theorem 2 (Finite-band attractor). *Let A solve (??) on a bounded C^∞ domain with periodic or homogeneous (Neumann/Dirichlet) boundary conditions. Then as $t \rightarrow \infty$ the Fourier spectrum of A concentrates in the band $\{k : \sigma(k) > 0\}$ and, moreover, for any $\delta > 0$ there exists $T_\delta < \infty$ such that for $t \geq T_\delta$ the spectral mass is contained in the annulus $\{k : ||k| - k_\star| < \delta\}$ up to a remainder that vanishes in L^2 as $t \rightarrow \infty$.*

Proof. **Step 1 (gradient flow, boundedness, and $\sigma \leq 0$ decay).** Define the Lyapunov functional

$$\mathcal{F}[A] = \int \left(-\frac{r}{2}|A|^2 + \frac{a}{2}|\nabla A|^2 + \frac{b}{2}|\nabla^2 A|^2 + \frac{\beta}{4}|A|^4 \right) dx.$$

Direct variation gives $\partial_t A = -\delta \mathcal{F} / \delta A^*$, hence

$$\frac{d\mathcal{F}}{dt} = - \int |\partial_t A|^2 dx \leq 0, \quad \mathcal{F}(t) \text{ is decreasing and bounded below.} \quad (47)$$

Thus A is uniformly bounded in H^2 and L^4 for all $t \geq 0$. Writing (??) in Fourier space ($\widehat{A}(k)$) gives

$$\partial_t \widehat{A}(k) = \sigma(k) \widehat{A}(k) - \beta \widehat{|A|^2 A}(k). \quad (48)$$

Fix $\eta > 0$ and let Q_η be the projector onto $\{k : \sigma(k) \leq -\eta\}$. Take L^2 inner product of $Q_\eta A$ with (??) and use Parseval:

$$\frac{1}{2} \frac{d}{dt} \|Q_\eta A\|_2^2 = \int_{\sigma \leq -\eta} \sigma(k) |\widehat{A}|^2 dk - \beta \Re \langle Q_\eta A, |A|^2 A \rangle \leq -\eta \|Q_\eta A\|_2^2 + C \|A\|_\infty^2 \|Q_\eta A\|_2^2,$$

where C is universal and $\|A\|_\infty$ is finite by Sobolev embedding from the H^2 bound. For late times $\|A\|_\infty$ is bounded uniformly; choosing η larger than $C\|A\|_\infty^2$ yields $\frac{d}{dt} \|Q_\eta A\|_2^2 \leq -\gamma \|Q_\eta A\|_2^2$ for some $\gamma > 0$, hence $\|Q_\eta A(t)\|_2 \rightarrow 0$ exponentially. Since $\eta > 0$ was arbitrary, the spectrum exits the set $\{\sigma \leq 0\}$ and concentrates in $\{\sigma > 0\}$.

Step 2 (selection near the peak $k_\star$). Let P_+ project onto $\{\sigma > 0\}$ and fix $\delta > 0$. On the compact set $\mathcal{A}_\delta := \{k : \sigma(k) > 0, ||k| - k_\star| \geq \delta\}$ one has

$$\sigma(k) \leq \sigma(k_\star) - c_\delta \quad \text{for some } c_\delta > 0$$

because $\sigma(k) = r - ak^2 - bk^4$ attains a strict maximum at $k_\star$ with $\sigma''(k_\star) = -4a < 0$ and is continuous. Define R_δ as the projector onto $\mathcal{A}_\delta$ and $S_\delta := P_+ - R_\delta$ (the thin annulus around $k_\star$). Taking the L^2 inner product of (48) with $R_\delta A$ gives

$$\frac{1}{2} \frac{d}{dt} \|R_\delta A\|_2^2 = \int_{\mathcal{A}_\delta} \sigma(k) |\widehat{A}|^2 dk - \beta \Re \langle R_\delta A, |A|^2 A \rangle \leq (\sigma_\star - c_\delta) \|R_\delta A\|_2^2 + C \|A\|_\infty^2 \|R_\delta A\|_2^2,$$

with $\sigma_\star := \sigma(k_\star)$. Meanwhile for the annulus part,

$$\frac{1}{2} \frac{d}{dt} \|S_\delta A\|_2^2 \geq (\sigma_\star - \varepsilon_\delta) \|S_\delta A\|_2^2 - C \|A\|_\infty^2 \|S_\delta A\|_2^2$$

with $\varepsilon_\delta \rightarrow 0$ as $\delta \rightarrow 0$ by continuity of σ . The cubic term saturates the total growth: $\frac{1}{2} \frac{d}{dt} \|P_+ A\|_2^2 = \int_{\sigma>0} \sigma |\widehat{A}|^2 - \beta \|A\|_4^4$, so in long time the net growth rate must vanish. Hence the only way to balance the linear advantage of S_δ over R_δ is that $\|R_\delta A(t)\|_2 \rightarrow 0$ as $t \rightarrow \infty$ while $\|S_\delta A\|_2$ saturates. Formally, for sufficiently small δ we have $(\sigma_\star - \varepsilon_\delta) - (\sigma_\star - c_\delta) = c_\delta - \varepsilon_\delta > 0$, so R_δ -modes are linearly underdriven relative to S_δ and are damped by the nonlinear sink. Grönwall's inequality with the uniform $\|A\|_\infty$ bound yields $\|R_\delta A(t)\|_2 \rightarrow 0$ as $t \rightarrow \infty$. Combining Step 1 and Step 2 gives the claim. $\square$

Corollary 3 (Shells and integer winding). If additionally $A = \rho e^{i\theta}$ with $\rho(x, t) \geq \rho_{\min} > 0$ on an open set for large t , then on any closed curve γ contained in that set,

$$\oint_\gamma \nabla \theta \cdot d\ell = 2\pi n, \quad n \in \mathbb{Z},$$

and the concentration of spectrum near $|k| \approx k_\star$ implies $|\nabla \theta| \approx k_\star$, so level sets organize into iso- $|k|$ shells; unwinding requires $\rho = 0$ somewhere on γ .

Proof. Where $\rho > 0$, θ is smooth and single-valued modulo 2π ; the integral of its gradient along a closed loop equals the net phase change, which must be an integer multiple of 2π by single-valuedness. Theorem 2 gives spectral concentration on a thin annulus, which is equivalent (via stationary phase) to $|\nabla \theta| \approx k_\star$ in the region where ρ is non-vanishing, hence shells are iso- $|k|$. A change of n requires a discontinuity in θ , which occurs only when ρ vanishes (phase slip). $\square$

A. Lyapunov Functional, Gradient Flow, and Condensation (Complete Proofs)

Setting. Let $\Omega = \mathbb{T}^d$ ($d \leq 3$) or a smooth bounded domain with periodic/Dirichlet/Neumann BCs. Assume $\psi(\cdot, t) \in H^2(\Omega)$ for all $t \geq 0$. Write $\Delta = \nabla \cdot \nabla$ and

$$g(\psi, \bar{\psi}) = \psi + \varepsilon e^{-\alpha|\psi|^2}, \quad \Theta[\psi] = -\frac{\Delta\psi}{g(\psi, \bar{\psi})}.$$

Definition 1 (Curvature Lyapunov).

$$E_0[\psi] = \int_{\Omega} (|\nabla\psi|^2 + |\Theta[\psi]|^2) dx, \quad E_{\text{pot}}[\psi] = \int_{\Omega} \left(\frac{\beta}{2} |\psi|^4 + \frac{\gamma}{2} |\psi|^2 \right) dx,$$

and $E := E_0 + E_{\text{pot}}$.

Lemma 2 (Gateaux derivative of E_0 w.r.t. $\bar{\psi}$). *Let $\eta \in H^2(\Omega)$ and consider the variation $\bar{\psi} \mapsto \bar{\psi} + \varepsilon \bar{\eta}$. Then*

$$\left. \frac{d}{d\varepsilon} E_0[\psi, \bar{\psi} + \varepsilon \bar{\eta}] \right|_{\varepsilon=0} = 2 \Re \int_{\Omega} \bar{\mu}_0 \eta dx, \quad (49)$$

with

$$\mu_0 = -\Delta\psi - \Delta\left(\frac{\Theta}{\bar{g}}\right) - \varepsilon \alpha e^{-\alpha|\psi|^2} \frac{\psi \Delta\psi \bar{\Theta}}{g^2}, \quad \bar{g} = \bar{\psi} + \varepsilon e^{-\alpha|\psi|^2}. \quad (50)$$

Proof. Write $E_0 = \int (|\nabla\psi|^2 + |\Theta|^2) dx$. Only $\bar{\psi}$ is varied. First, $\frac{d}{d\varepsilon} \int |\nabla\psi|^2 dx = 2 \Re \int \nabla\psi \cdot \nabla \bar{\eta} dx = -2 \Re \int \Delta\psi \bar{\eta} dx$ by integration by parts (BCs legitimate by hypothesis), contributing $-\Delta\psi$ to μ_0 .

Second, $F[\psi, \bar{\psi}] := \int |\Theta|^2 dx = \int \Theta \bar{\Theta} dx$ gives

$$\delta_{\bar{\psi}} F = \int \Theta \delta \bar{\Theta} dx = \int \Theta \left[-\frac{\Delta \bar{\eta}}{\bar{g}} - \frac{\Delta \bar{\psi}}{\bar{g}^2} \partial_{\bar{\psi}} \bar{g} \bar{\eta} \right] dx,$$

since $\bar{\Theta} = -\Delta \bar{\psi} / \bar{g}$ and $\delta(1/\bar{g}) = -\bar{g}^{-2} \delta \bar{g}$. Here $\partial_{\bar{\psi}} \bar{g} = 1 - \alpha \varepsilon e^{-\alpha|\psi|^2} \psi$. Integrate by parts the first term:

$$\int \Theta \left(-\frac{\Delta \bar{\eta}}{\bar{g}} \right) dx = \int \nabla \left(\frac{\Theta}{\bar{g}} \right) \cdot \nabla \bar{\eta} dx = - \int \Delta \left(\frac{\Theta}{\bar{g}} \right) \bar{\eta} dx.$$

Thus

$$\delta_{\bar{\psi}} F = - \int \Delta \left(\frac{\Theta}{\bar{g}} \right) \bar{\eta} dx - \int \Theta \frac{\Delta \bar{\psi}}{\bar{g}^2} (1 - \alpha \varepsilon e^{-\alpha|\psi|^2} \psi) \bar{\eta} dx.$$

Take real part and recast in the $\int \bar{\mu} \eta$ form. Using $\Theta = -\Delta\psi/g$, we have $\Theta \Delta \bar{\psi} / \bar{g}^2 = -(\Delta\psi \Delta \bar{\psi}) / (g \bar{g}^2)$. The factor 1 in the parenthesis yields a purely real contribution that cancels against its complex conjugate in the $2 \Re(\cdot)$ form when assigned to μ vs. $\bar{\mu}$. The only surviving complex (non-self-conjugate) contribution comes from the $(-\alpha \varepsilon e^{-\alpha|\psi|^2} \psi)$ piece, producing

$$- \left(-\alpha \varepsilon e^{-\alpha|\psi|^2} \psi \right) \Theta \frac{\Delta \bar{\psi}}{\bar{g}^2} = -\varepsilon \alpha e^{-\alpha|\psi|^2} \frac{\psi \Delta\psi \bar{\Theta}}{g^2}.$$

Collecting with the $-\Delta(\Theta/\bar{g})$ term gives (50). Finally, (49) follows by the standard identification $\delta E_0 = 2 \Re \int \bar{\mu}_0 \eta dx$. $\square$

Lemma 3 (Full chemical potential). *For $E = E_0 + E_{\text{pot}}$,*

$$\mu = \frac{\delta E}{\delta \psi} = \mu_0 + \beta |\psi|^2 \psi + \frac{\gamma}{2} \psi,$$

with μ_0 given by (50).

Proof. Immediate from Lemma 2 and $\delta_{\bar{\psi}} \int (\beta |\psi|^4 / 2) dx = \int \beta |\psi|^2 \psi \bar{\eta} dx$, $\delta_{\bar{\psi}} \int (\gamma |\psi|^2 / 2) dx = \int (\gamma / 2) \psi \bar{\eta} dx$. $\square$

Theorem 4 (Mixed-metric gradient flow: strict Lyapunov descent). *Consider*

$$\partial_t \psi = -\kappa \mu + D \Delta \mu, \quad \mu = \frac{\delta E}{\delta \psi}, \quad \kappa > 0, D \geq 0. \quad (51)$$

Then for all classical solutions with the stated BCs,

$$\frac{d}{dt} E[\psi(t)] = \Re \int_{\Omega} \bar{\mu} \partial_t \psi dx = -\kappa \int_{\Omega} |\mu|^2 dx - D \int_{\Omega} |\nabla \mu|^2 dx \leq 0. \quad (52)$$

Moreover, $E(t)$ is strictly decreasing unless $\mu \equiv 0$.

Proof. By the chain rule for Gâteaux-differentiable E along a smooth trajectory, $\dot{E} = \Re \int \bar{\mu} \partial_t \psi dx$. Insert (51):

$$\dot{E} = -\kappa \int |\mu|^2 dx + D \Re \int \bar{\mu} \Delta \mu dx = -\kappa \int |\mu|^2 dx - D \int |\nabla \mu|^2 dx,$$

the last equality by integration by parts (BCs). Nonnegativity is immediate. If $\mu \not\equiv 0$, at least one of $\int |\mu|^2$, $\int |\nabla \mu|^2$ is > 0 , hence $\dot{E} < 0$. $\square$

Corollary 5 (Allen–Cahn and Cahn–Hilliard rails). $D = 0$ yields the L^2 -gradient flow with $\dot{E} = -\kappa \int |\mu|^2 \leq 0$. $\kappa = 0, D > 0$ yields the H^{-1} -gradient flow with $\dot{E} = -D \int |\nabla \mu|^2 \leq 0$. In both cases, $\dot{E} = 0$ iff $\mu \equiv 0$.

Spectral quantities. On $\Omega = \mathbb{T}^d$, write the Fourier coefficients $\hat{\psi}(k, t)$, $k \in \mathbb{Z}^d$, and the normalized spectral power

$$P(k, t) := \frac{|\hat{\psi}(k, t)|^2}{\sum_{q \in \mathbb{Z}^d} |\hat{\psi}(q, t)|^2}, \quad H_k(t) := - \sum_k P(k, t) \log P(k, t).$$

For $\epsilon > 0$, denote by $\mathcal{N}_\epsilon(S)$ the ϵ -neighborhood of a set $S \subset \mathbb{R}^d$.

Assumption 6 (Band coercivity & finite pass-band). *There exist constants $\kappa, \theta > 0$, $\gamma \in \mathbb{R}$ with $\gamma^2 < 4\kappa\theta$ and $\lambda > 0$ such that the linearized symbol*

$$\mathcal{L}(k) := \kappa(\omega^2 - k^2)^2 - \gamma k^2(\omega^2 - k^2) + \theta k^4 - \lambda$$

admits a finite pass-band centered at $|k| \approx k_$ and a coercive complement: there exist $c_{\text{out}}, K > 0$ and an annulus*

$$\mathcal{A} = \{ k : ||k| - k_*| \leq K \}$$

with

$$E_0[\psi] \geq c_{\text{out}} \sum_{k \notin \mathcal{A}} (1 + |k|^4) |\hat{\psi}(k)|^2 \quad \text{for all } \psi \in H^2(\Omega). \quad (53)$$

Lemma 4 (Uniform bounds and precompactness). *Let ψ solve (51) on $\mathbb{T}^d$ and suppose $E[\psi(0)] < \infty$. Then $E(t)$ is nonincreasing and bounded below. Under Assumption 6, $\{\psi(\cdot, t) : t \geq 0\}$ is bounded in $H^2(\Omega)$ and precompact in $H^s(\Omega)$ for any $s < 2$.*

Proof. The first statement is Theorem 4. By Definition 1 and (53),

$$E(t) \geq \int |\nabla \psi|^2 dx + c_{\text{out}} \sum_{k \notin \mathcal{A}} (1 + |k|^4) |\widehat{\psi}(k)|^2 + \frac{\beta}{2} \|\psi\|_{L^4}^4 + \frac{\gamma}{2} \|\psi\|_{L^2}^2.$$

Thus $E(t) \leq E(0)$ implies uniform control of the H^2 -tail outside $\mathcal{A}$ and the H^1 -norm globally; within $\mathcal{A}$ there are finitely many modes on the torus. Hence $\|\psi(\cdot, t)\|_{H^2}$ is uniformly bounded. Rellich–Kondrachov gives precompactness in H^s , $s < 2$. $\square$

Proposition 2 (ω -limit set and stationarity). *Under Lemma 4, there exists a nonempty compact ω -limit set $\omega(\psi_0) \subset H^s(\Omega)$, $s < 2$, such that every $\phi \in \omega(\psi_0)$ satisfies $\mu[\phi] = 0$ (i.e. it is a steady state of (51)).*

Proof. Pick $t_n \rightarrow \infty$ with $\psi(\cdot, t_n)$ convergent in H^s . By Theorem 4, $\int_0^\infty \|\mu\|_{L^2}^2 dt < \infty$ (or $\int_0^\infty \|\nabla \mu\|_{L^2}^2 dt < \infty$ if $\kappa = 0$), so a subsequence satisfies $\|\mu(\cdot, t_n)\|_{L^2} \rightarrow 0$. Continuity of $\mu : H^2 \rightarrow L^2$ yields $\mu[\phi] = 0$ for the limit ϕ . $\square$

Theorem 7 (Condensation under band coercivity). *Let Assumption 6 hold and ψ solve (51) with $E(0) < \infty$. Fix $\epsilon > 0$ and let $\mathcal{A}_\epsilon = \mathcal{N}_\epsilon(\mathcal{A})$. Then*

$$\lim_{t \rightarrow \infty} \sum_{k \notin \mathcal{A}_\epsilon} |\widehat{\psi}(k, t)|^2 = 0. \quad (54)$$

Consequently, for the normalized spectrum $P(k, t)$,

$$\forall \delta > 0 \exists T(\delta) \text{ s.t. } t \geq T(\delta) \Rightarrow \sum_{k \notin \mathcal{A}_\epsilon} P(k, t) \leq \delta,$$

and the Shannon spectral entropy $H_k(t)$ converges to a finite limit H_∞ .

Proof. By Lemma 4 and Proposition 2, every sequence $t_n \rightarrow \infty$ has a subsequence t_{n_j} such that $\psi(\cdot, t_{n_j}) \rightarrow \phi$ in H^s , $s < 2$, with $\mu[\phi] = 0$. Any such ϕ must satisfy $\widehat{\phi}(k) = 0$ for $k \notin \mathcal{A}$: if some $k_0 \notin \mathcal{A}$ had $|\widehat{\phi}(k_0)| > 0$, then the quadratic part of E (see (53)) would strictly decrease under moving that mass into $\mathcal{A}$, contradicting stationarity of ϕ in the finite Fourier subspace spanned by $\{k_0\} \cup \mathcal{A}$. Let Π_{out} project onto $\mathbb{Z}^d \setminus \mathcal{A}_\epsilon$. Then $\|\Pi_{\text{out}} \psi(\cdot, t_{n_j})\|_{L^2} \rightarrow 0$; since every divergent sequence has such a subsequence, $\lim_{t \rightarrow \infty} \|\Pi_{\text{out}} \psi(\cdot, t)\|_{L^2} = 0$, proving (54). The claim for $P(k, t)$ follows by normalization; tightness on the finite set $\mathcal{A}_\epsilon \cap \mathbb{Z}^d$ yields convergence of $H_k(t)$. $\square$

Corollary 8 (Finite coherent support). *On $\mathbb{T}^d$, $\mathcal{A} \cap \mathbb{Z}^d$ is finite. There exists a finite $K^* \subset \mathbb{Z}^d$ and, for all small $\epsilon > 0$,*

$$\lim_{t \rightarrow \infty} \sum_{k \notin K^*} P(k, t) = 0.$$

Thus, asymptotically, the spectral mass concentrates on finitely many stationary peaks.

Proof. $\mathcal{A}$ is compact; $\mathcal{A} \cap \mathbb{Z}^d$ is finite. Take ϵ small so that $\mathcal{A}_\epsilon \cap \mathbb{Z}^d = K^*$ and apply Theorem 7. $\square$

B Computational Review Note

Several numerical and logical assessments in this paper were assisted by ChatGPT, an AI language model developed by OpenAI. The model was used to check order-of-magnitude estimates, structural coherence, and consistency with existing scientific literature. All factual claims remain the responsibility of the author.